Topic: Cosine and sine series of piecewise functions

Question: Find the Fourier sine series representation of the piecewise function on $0 \leq x \leq L$.

$$f(x) = \begin{cases} -L & 0 \leq x \leq \frac{L}{2} \\ L & \frac{L}{2} < x \leq L \end{cases}$$

Answer choices:

A $$f(x) = \frac{4L}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = \frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left(2\cos\left(\frac{n\pi}{2}\right) - 1 + (-1)^{n+1}\right) \sin\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = -\frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left(\cos\left(\frac{n\pi}{2}\right) + 1 + (-1)^{n+1}\right) \sin\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = \frac{L}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} \left((-1)^n - \sin\left(\frac{n\pi}{2}\right)\right) \sin\left(\frac{n\pi x}{L}\right)$$

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Solution: B

We want to build the Fourier sine series representation, so we'll start by finding the odd extension of the piecewise function. Because the original piecewise function $f(x)$ is defined in two pieces, its odd extension $g(x)$ will be defined in four pieces.

$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} L & \frac{L}{2} < x \le L \\ -L & 0 \le x \le \frac{L}{2} \\ L & -\frac{L}{2} \le x < 0 \\ -L & -L \le x < -\frac{L}{2} \end{cases}$$

To find the Fourier sine series representation, we'll calculate B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^{\frac{L}{2}} (-L)\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{\frac{L}{2}}^L L\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -2\int_0^{\frac{L}{2}} \sin\left(\frac{n\pi x}{L}\right)\,dx + 2\int_{\frac{L}{2}}^L \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} - \frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^L$$

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$$B_n = \frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{2L}{n\pi}\cos(0) - \left(\frac{2L}{n\pi}\cos(n\pi) - \frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right)\right)$$

$$B_n = \frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{2L}{n\pi} - \frac{2L}{n\pi}\cos(n\pi) + \frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right)$$

$$B_n = \frac{4L}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{2L}{n\pi}(1 + (-1)^n)$$

Then the Fourier sine series representation of the piecewise function on $0 \le x \le L$ is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\left(2\cos\left(\frac{n\pi}{2}\right) - 1 + (-1)^{n+1}\right)\sin\left(\frac{n\pi x}{L}\right)$$

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Topic: Cosine and sine series of piecewise functions

Question: Find the Fourier sine series representation of the piecewise function on $0 \leq x \leq L$.

$$f(x) = \begin{cases} L - x & 0 \leq x \leq \frac{L}{2} \\ x - L & \frac{L}{2} < x \leq L \end{cases}$$

Answer choices:

A $$f(x) = \frac{L}{2\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left((-1)^n \frac{L^2 + n^2}{n^2} - \sin\left(\frac{n\pi}{2}\right) \right) \sin\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = \frac{L}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin\left(\frac{n\pi x}{L}\right) + (-1)^{n+1} \cos\left(\frac{n\pi x}{L}\right) \right)$$

C $$f(x) = \frac{L}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left(1 - \cos\left(\frac{n\pi}{2}\right) - 2\sin\left(\frac{n\pi}{2}\right) \right) \sin\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = \frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left(1 - \cos\left(\frac{n\pi}{2}\right) - \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \right) \sin\left(\frac{n\pi x}{L}\right)$$

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Solution: D

We want to build the Fourier sine series representation, so we'll start by finding the odd extension of the piecewise function. Because the original piecewise function $f(x)$ is defined in two pieces, its odd extension $g(x)$ will be defined in four pieces.

$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} x - L & \frac{L}{2} < x \le L \\ L - x & 0 \le x \le \frac{L}{2} \\ -L - x & -\frac{L}{2} \le x < 0 \\ x + L & -L \le x < -\frac{L}{2} \end{cases}$$

To find the Fourier sine series representation, we'll calculate B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^{\frac{L}{2}} (L-x)\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{\frac{L}{2}}^L (x-L)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^{\frac{L}{2}} L\sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{2}{L}\int_0^{\frac{L}{2}} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$+\frac{2}{L}\int_{\frac{L}{2}}^L x\sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{2}{L}\int_{\frac{L}{2}}^L L\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = 2\int_0^{\frac{L}{2}} \sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{2}{L}\int_0^{\frac{L}{2}} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$+\frac{2}{L}\int_{\frac{L}{2}}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx - 2\int_{\frac{L}{2}}^{L} \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{2}} + \frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^{L}$$

$$-\frac{2}{L}\int_0^{\frac{L}{2}} x\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{\frac{L}{2}}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{n\pi}\cos 0 + \frac{2L}{n\pi}\cos n\pi - \frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right)$$

$$-\frac{2}{L}\int_0^{\frac{L}{2}} x\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{\frac{L}{2}}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{4L}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{n\pi}(1+(-1)^n) - \frac{2}{L}\int_0^{\frac{L}{2}} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$+\frac{2}{L}\int_{\frac{L}{2}}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$\int x\sin\left(\frac{n\pi x}{L}\right)\,dx = -\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)dx$$

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$$\int x \sin\left(\frac{n\pi x}{L}\right)\, dx = -\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

Plugging this back into the equation for B_n, we get

$$B_n = -\frac{4L}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{n\pi}(1+(-1)^n)$$

$$-\frac{2}{L}\left[-\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^{\frac{L}{2}}$$

$$+\frac{2}{L}\left[-\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_{\frac{L}{2}}^{L}$$

$$B_n = -\frac{4L}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{n\pi}(1+(-1)^n)$$

$$+\left[\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^{\frac{L}{2}}$$

$$+\left[-\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_{\frac{L}{2}}^{L}$$

$$B_n = -\frac{4L}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{n\pi}(1+(-1)^n) + \frac{2L}{2n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)$$

$$-\frac{2L}{n\pi}\cos n\pi + \frac{2L}{(n\pi)^2}\sin n\pi + \frac{2L}{2n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)$$

$$B_n = -\frac{4L}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2L}{n\pi} + \frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{4L}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)$$

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$$B_n = \frac{2L}{n\pi} - \frac{2L}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{4L}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)$$

Then the Fourier sine series representation of the piecewise function on $0 \leq x \leq L$ is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\left(1 - \cos\left(\frac{n\pi}{2}\right) - \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right)\sin\left(\frac{n\pi x}{L}\right)$$

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Topic: Cosine and sine series of piecewise functions

Question: Find the Fourier cosine series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} L & -L \leq x < -\frac{L}{2} \\ x^2 & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ L & \frac{L}{2} < x \leq L \end{cases}$$

Answer choices:

A $$f(x) = \frac{L^2}{8} - \frac{L}{2\pi^3} \sum_{n=1}^{\infty} \frac{4L}{n^3} \left(2 \sin\left(\frac{n\pi}{2}\right) - n\pi \cos\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = L + \frac{L^3}{9} + \frac{L}{\pi^2} \sum_{n=1}^{\infty} \frac{n^2}{\pi^2 - n} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = \frac{L^2}{24} + \frac{L}{2} + \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{L(n\pi)^2(L-4) - 8L^2}{(n\pi)^3} \sin\left(\frac{n\pi}{2}\right) + \frac{4L^2}{(n\pi)^2} \cos\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = \frac{L^2}{24} - L - \frac{L}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^2} \left((L^2 - n\pi) \sin\left(\frac{n\pi}{2}\right) - \sin\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

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Solution: C

The function is even because $f(-x) = f(x)$.

$$f(-x) = \begin{cases} L & -L \leq x < -\frac{L}{2} \\ x^2 & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ L & \frac{L}{2} < x \leq L \end{cases} = f(x)$$

To find the Fourier cosine series, we'll calculate A_0 and A_n. For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^{\frac{L}{2}} x^2\ dx + \frac{1}{L}\int_{\frac{L}{2}}^L L\ dx$$

$$A_0 = \frac{1}{L}\left(\frac{x^3}{3}\right)\Bigg|_0^{\frac{L}{2}} + \frac{1}{L}(Lx)\Bigg|_{\frac{L}{2}}^L$$

$$A_0 = \frac{x^3}{3L}\Bigg|_0^{\frac{L}{2}} + x\Bigg|_{\frac{L}{2}}^L$$

$$A_0 = \frac{\left(\frac{L}{2}\right)^3}{3L} - \frac{0^3}{3L} + L - \frac{L}{2}$$

$$A_0 = \frac{L^2}{24} + \frac{L}{2}$$

And for A_n we get

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$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\int_0^{\frac{L}{2}} x^2\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{\frac{L}{2}}^L L\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\int_0^{\frac{L}{2}} x^2\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{\frac{L}{2}}^L$$

For the remaining integral, use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^{\frac{L}{2}} - \frac{4}{n\pi}\int_0^{\frac{L}{2}} x\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{\frac{L}{2}}^L$$

Use integration by parts again with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = \frac{2x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^{\frac{L}{2}} + \frac{4Lx}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^{\frac{L}{2}} - \frac{4L}{(n\pi)^2}\int_0^{\frac{L}{2}}\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$+\frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{\frac{L}{2}}^L$$

$$A_n = \frac{2x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^{\frac{L}{2}} + \frac{4Lx}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^{\frac{L}{2}} - \frac{4L^2}{(n\pi)^3}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^{\frac{L}{2}}$$

$$+\frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{\frac{L}{2}}^L$$

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Now evaluate over each interval.

$$A_n = \frac{L^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{L^2}{2n\pi}\sin(0) + \frac{2L^2}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right) - 0 + \frac{4L^2}{(n\pi)^3}\sin\left(\frac{n\pi}{2}\right)$$

$$+\frac{4L^2}{(n\pi)^3}\sin(0) + \frac{2L}{n\pi}\sin(n\pi) - \frac{2L}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

$$A_n = \frac{L^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{2L^2}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right) - \frac{4L^2}{(n\pi)^3}\sin\left(\frac{n\pi}{2}\right) - \frac{2L}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

$$A_n = \left(\frac{L^2}{2n\pi} - \frac{4L^2}{(n\pi)^3} - \frac{2L}{n\pi}\right)\sin\left(\frac{n\pi}{2}\right) + \frac{2L^2}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right)$$

$$A_n = \left(\frac{L^2 - 4L}{2n\pi} - \frac{4L^2}{(n\pi)^3}\right)\sin\left(\frac{n\pi}{2}\right) + \frac{2L^2}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right)$$

$$A_n = \frac{L(n\pi)^2(L-4) - 8L^2}{2(n\pi)^3}\sin\left(\frac{n\pi}{2}\right) + \frac{2L^2}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right)$$

Then the Fourier cosine series representation of the piecewise function on $0 \le x \le L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{L^2}{24} + \frac{L}{2}$$

$$+\frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{L(n\pi)^2(L-4) - 8L^2}{(n\pi)^3}\sin\left(\frac{n\pi}{2}\right) + \frac{4L^2}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right)\right)\cos\left(\frac{n\pi x}{L}\right)$$

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